

Kiara M. Ascencion Arevalo

Ontology Engineer | Knowledge Graph Engineer
M. Sc. Information Systems

PERSONAL INFORMATION

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Telephone +49 176 66082461
Nationality Peruvian

LANGUAGES

English ★★★★★
Spanish ★★★★★
German ★★★★★
Chinese ★★★★★☆

SKILLS

Knowledge Graph	★★★★★	Ontology Engineering	★★★★★
RDF	★★★★★	Protegé	★★★★☆
RDFS, OWL	★★★★★	GraphDB	★★★★☆
SPARQL	★★★★★	RDFlib	★★★★☆
RML, R2RML	★★★☆☆	NLP, LLM	★★★★☆
SHACL	★★★☆☆	Python	★★★★☆

EDUCATION

2018 – 2021 **Friedrich-Alexander-University Erlangen-Nuremberg**, Germany
M.Sc. in International Information Systems, School of Business and Economics

2013 – 2017 **Feng Chia University (FCU)**, Taiwan
B.B.A. in International Business Administration, College of Business

WORKING EXPERIENCE

1/2026 – Present **metaphacts GmbH (part of Digital Science GmbH)**, Germany, Germany
Technical Consultant – Semantic Web & Knowledge Graph Solutions

7/2021 – 12/2025 **Technische Hochschule Nürnberg Georg Simon Ohm**, Germany
Associate Researcher

- ❖ Collaborate on scientific papers and contribute to academic publishing in the semantic technologies domain
- ❖ Develop data acquisition and analysis processes with Python to improve data insights
- ❖ Engineer ontologies for comprehensive knowledge structuring focused on technological and market-based information using OWL-based tools
- ❖ Streamline ontology development processes through the selection and application of NLP methods
- ❖ Conduct evaluation of processes and outcomes to ensure research quality

7/2021 – 12/2025 **Fraunhofer-Institut für Integrierte Schaltungen IIS**, Germany
External Consultant

- ❖ Conceptualize and lead the execution of semantic web-based research initiatives in collaboration with industry and academia
- ❖ Analyze customer and stakeholder requirements to define engineering data exchange processes and support interoperable architecture design
- ❖ Project management and coordination applying both traditional and agile methodologies
- ❖ Develop and manage ontologies and semantic models with a focus on interoperability and industry standards

6/2019 – 6/2021	Fraunhofer-Institut für Integrierte Schaltungen IIS, Germany <i>Student Research Assistant – part-time</i> ❖ Utilize semantic web technologies (RDF, SPARQL, OWL) for knowledge modeling and data interpretation ❖ Manage relational database operations and maintenance for ETL processes ❖ Conduct data migration from RDB to triple stores and optimized ETL with Python, leveraging AI and NLP methods ❖ Develop ontologies with Protégé and integrated NLP methods for information extraction, improving data usability and analysis
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RESEARCH PROJECTS

3/2024 – 12/2025	DuraFuelCell: Multi-scale investigation for the development of durable and efficient hydrogen fuel cell systems for mobile and stationary applications <i>Project Lead, Ontology Engineer</i> Funded by the German Research Foundation (DFG), the joint project focuses on developing long-lasting, efficient hydrogen fuel cell systems for both mobile and stationary applications. Within this impulse, the 'Data-based trend and scenario research' unit, aims to harness Semantic Web technologies to construct an AI-based knowledge database that serves as a central platform to structure international research, facilitate knowledge exchange, and improve efficiency in hydrogen fuel cell technology research.
7/2021 – 12/2025	Atlant-H – Artificial Market Intelligence in the context of hydrogen <i>Project Lead, Ontology engineer, Knowledge Graph Engineer</i> The project aims to create a graph-based analytical tool to enhance fuel cell research within the hydrogen economy. By using NLP and semantic web technologies to process and analyze vast amounts of textual data, the goal is to generate predictions and offer a comprehensive, data-driven view of the hydrogen economy through customizable dashboards, targeting advancements in fuel cell technologies.
1/2022 – 12/2022	Data4Leads – Automated text analyses for lead generation to support new business development <i>Project Lead, Knowledge Graph Engineer</i> The aim of the preliminary research project "Data4Lead" was to automatically identify relevant customer contacts ("leads") from unstructured text data using a new, tailored NLP approach. The aim was to transfer an experience-based, heuristic approach to lead identification into an automatable process.

PUBLICATIONS

11/2023	Ascencion Arevalo, K.M. , Neunsinger, C., Zimmermann, R., Blum, R., Weakly, K.: <i>HOLY: An Ontology Covering the Hydrogen Market</i> . In International Semantic Web Conference 2023 (ISWC), Greece. DOI: 10.1007/978-3-031-47243-5_1
02/2025	Ascencion Arevalo, K. M. , Ambre, S., Dorsch, R.: <i>AutOnto: Towards A Semi-Automated Ontology Engineering Methodology</i> . In Knowledge Graph and Semantic Web Conference 2024 (KGSWC), DOI: 10.1007/978-3-031-81221-7_16

PROFESSIONAL CERTIFICATION COURSES

2/2024	Professional Scrum Master I Scrum.org
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